Electrostatic Potential and Field of a Charged Sphere: Spherical Harmonics Approach

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Abstract

The electrostatic potential and field produced throughout space by a charged spherical surface are derived by employing the spherical harmonics expansion of $\frac{1}{|\mathbf{r}-\mathbf{r}'|}$, a term proportional to the Green function of the three-dimensional Laplacian. Once the potential is determined, the electric field is obtained by taking its negative gradient. This approach provides an efficient alternative to direct evaluation of the potential integral.

1 Introduction

The determination of the electrostatic potential $\Phi(\mathbf{r})$ generated by a charge distribution is a central problem in electromagnetic theory. In this note, we consider a hollow sphere of radius r_0 with a non-uniform surface charge density given by $\sigma(\mathbf{r}) = \sigma(\theta) = \sigma_0 \cos\theta$. Our objective is to derive the potential everywhere in space via the integral method.

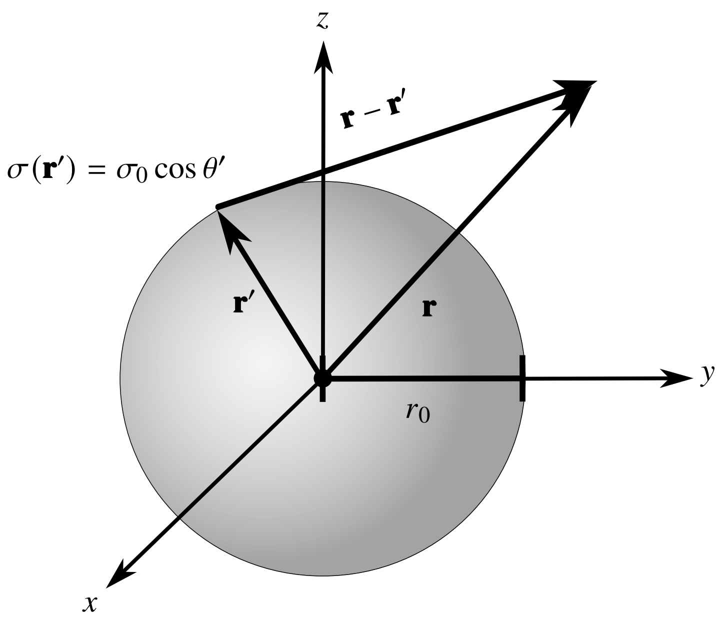

Figure 1: *Physical system.* Spherical surface with charge density $\sigma(\mathbf{r}') = \sigma_0 \cos\theta'$. The use of the prime notation will be explained in the next section.

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2 Potential Integral Formulation

The electric potential $\Phi(\mathbf{r})$ generated by a closed surface is given by the integral formula:

$$\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \oint_S \frac{\sigma(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, da' , \tag{1}$$

where $\mathbf{r}$ is the observation point, i.e., the location in space where the electric potential is being evaluated. By convention, the primed vector $\mathbf{r}'$ represents the source. The prime denotes what varies in the integral, i.e., we can see $\mathbf{r}'$ as a vector that sweeps across the surface.

We consider each component in Eq. (1) separately. The differential surface element for a sphere of radius r_0 is expressed in spherical coordinates as:

$$da' = r_0^2 \sin\theta'\, d\theta'\, d\phi' .$$

For a full sphere, the angular coordinates range over:

$$\begin{aligned} 0 &\le \theta' \le \pi \\ 0 &\le \phi' < 2\pi . \end{aligned}$$

In addition, we recall the charge density introduced earlier:

$$\sigma(\mathbf{r}') = \sigma_0 \cos\theta' .$$

Finally, to evaluate the integral for a general observation point, we invoke the spherical harmonics expansion of the term $\dfrac{1}{|\mathbf{r} - \mathbf{r}'|}$:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \begin{cases} 4\pi \displaystyle\sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \frac{r^l}{r'^{l+1}} & \text{if} \quad r < r' \\ 4\pi \displaystyle\sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \frac{r'^l}{r^{l+1}} & \text{if} \quad r > r' . \end{cases} \tag{2}$$

3 Application of the Spherical Harmonics Expansion

Exploiting the symmetry between the two expressions in Eq. (2), it is common to use a compact form of the expansion by introducing the symbols $r_<$ and $r_>$ as follows:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) , \tag{3}$$

where $r_<$ is the smaller of r and r', and $r_>$ is the larger of these two. This notation allows us to evaluate the integral simultaneously in both regions of interest (inside and outside the sphere). Furthermore, although r' represents the radial integration variable, in this problem we are dealing with a charged spherical surface, so r' is constant, and its value is the radius of the sphere, yielding:

$$r' = r_0 .$$

Substituting the specified charge density, the differential surface element, and the expansion of Eq. (3) into the integral of Eq. (1), we obtain:

$$\Phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int_0^{2\pi} \int_0^{\pi} \sigma_0 \cos\theta' \left[4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \right] r_0^2 \sin\theta'\, d\theta'\, d\phi' .$$

Canceling the 4π factor and extracting the constants from the integral yields:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{\varepsilon_0} \int_0^{2\pi} \int_0^{\pi} \cos\theta' \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \sin\theta' \, d\theta' \, d\phi' \, .$$

For this expansion, it is valid to interchange sum and integral as follows:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{\varepsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) \int_0^{2\pi} \int_0^{\pi} \cos\theta' Y_{lm}^*(\theta', \phi') \sin\theta' \, d\theta' \, d\phi' \, . \tag{4}$$

Note that, as established earlier, we have treated the unprimed variables as constants with respect to the integration.

It is convenient for what comes next to express $\cos\theta'$ in terms of spherical harmonics. For this function, we recall:

$$Y_{10}(\theta', \phi') = \sqrt{\frac{3}{4\pi}} \cos\theta' \, ,$$

where $\sqrt{\frac{3}{4\pi}}$ is a normalization constant. From this relation it follows that:

$$\cos\theta' = \sqrt{\frac{4\pi}{3}} Y_{10}(\theta', \phi') \, . \tag{5}$$

We substitute in Eq. (4):

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{\varepsilon_0} \sqrt{\frac{4\pi}{3}} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) \int_0^{2\pi} \int_0^{\pi} Y_{lm}^*(\theta', \phi') Y_{10}(\theta', \phi') \sin\theta' \, d\theta' \, d\phi' \, .$$

To evaluate the double integral, we employ the orthogonality relation of the spherical harmonics:

$$\int_0^{2\pi} \int_0^{\pi} Y_{lm}^*(\theta', \phi') Y_{l'm'}(\theta', \phi') \sin\theta' \, d\theta' \, d\phi' = \delta_{ll'} \delta_{mm'} \, ,$$

which reduces our double integral to a product of Kronecker deltas. Identifying $l' = 1$ and $m' = 0$ for this case, we obtain:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{\varepsilon_0} \sqrt{\frac{4\pi}{3}} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) \delta_{l1} \delta_{m0} \, .$$

Due to the definition of the Kronecker delta, the double sum collapses. All terms in the sum over index m are 0, except the one with $m = 0$. Similarly, in the sum over the index l, all terms are 0, except the one with $l = 1$. This yields:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{\varepsilon_0} \sqrt{\frac{4\pi}{3}} \frac{1}{3} \frac{r_<}{r_>^2} Y_{10}(\theta, \phi) \, .$$

Applying the relation of Eq. (5), the result is:

$$\therefore \quad \Phi(\mathbf{r}) = \frac{\sigma_0 r_0^2}{3\varepsilon_0} \frac{r_<}{r_>^2} \cos\theta \, . \tag{6}$$

4 Evaluation of the Potential in Different Regions

From Eq. (6), we analyze the potential inside and outside the sphere. Recalling the definition of the symbols $r_<$ and $r_>$, together with the fact that $r' = r_0$, it is straightforward to treat both regions.

4.1 Inside the Sphere ($r < r_0$)

$$r_< = r \quad \& \quad r_> = r_0 \,.$$

This implies that:

$$\Phi(\mathbf{r}) = \frac{\sigma_0}{3\varepsilon_0} r \cos\theta \,.$$

Note that, from spherical coordinates, we have the relation $r\cos\theta = z$. Changing to Cartesian coordinates is useful to analyze the potential in a simpler manner. Furthermore, this strategy simplifies the calculation of the electric field.

$$\Phi(\mathbf{r}) = \frac{\sigma_0}{3\varepsilon_0} z \,. \tag{7}$$

4.2 Outside the Sphere ($r > r_0$)

$$r_< = r_0 \quad \& \quad r_> = r \,.$$

Consequently:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2} \cos\theta \,. \tag{8}$$

4.3 Electrostatic Potential Everywhere in Space

Due to continuity of the potential, the values inside and outside the sphere coincide at $r = r_0$. Therefore, it is convenient to take the simpler form.

Combining the expressions of Eqs. (7) and (8), we obtain:

$$\Phi(\mathbf{r}) = \begin{cases} \dfrac{\sigma_0}{3\varepsilon_0} z & \text{if} \quad r \leq r_0 \\ \dfrac{\sigma_0 r_0^3}{3\varepsilon_0 r^2} \cos\theta & \text{if} \quad r > r_0 \,. \end{cases} \tag{9}$$

5 Derivation of the Electrostatic Field

With the electrostatic potential determined in Eq. (9), the calculation of the field is done via negative gradient.

$$\mathbf{E}(\mathbf{r}) = -\boldsymbol{\nabla}\Phi(\mathbf{r}) \,.$$

Since the potential behaves differently inside and outside the sphere, the electric field is evaluated separately in each region.

5.1 Inside the Sphere ($r < r_0$)

From Eq. (7) we know that:

$$\Phi(\mathbf{r}) = \frac{\sigma_0}{3\varepsilon_0} z \,.$$

Applying the negative gradient in cartesian coordinates yields:

$$-\boldsymbol{\nabla}\Phi(\mathbf{r}) = -\boldsymbol{\nabla}\frac{\sigma_0}{3\varepsilon_0} z$$

$$\mathbf{E}(\mathbf{r}) = -\frac{\partial}{\partial z}\left(\frac{\sigma_0}{3\varepsilon_0} z\right)\hat{\mathbf{k}}$$

$$\therefore \quad \mathbf{E}(\mathbf{r}) = -\frac{\sigma_0}{3\varepsilon_0}\,\hat{\mathbf{k}}\,. \tag{10}$$

5.2 Outside the Sphere ($r > r_0$)

From Eq. (8) we have:

$$\Phi(\mathbf{r}) = \frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\,.$$

Applying the negative gradient yields:

$$-\boldsymbol{\nabla}\Phi(\mathbf{r}) = -\boldsymbol{\nabla}\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\,.$$

We recall the gradient operator in spherical coordinates:

$$\boldsymbol{\nabla} = \hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\mathbf{e}}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}$$

$$\mathbf{E}(\mathbf{r}) = -\left(\hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\mathbf{e}}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\right)\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\,. \tag{11}$$

It is convenient to calculate each derivative separately:

$$\frac{\partial}{\partial r}\left(\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\right) = -\frac{2\sigma_0 r_0^3}{3\varepsilon_0 r^3}\cos\theta \tag{12}$$

$$\frac{\partial}{\partial\theta}\left(\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\right) = -\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\sin\theta$$

$$\frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\right) = -\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^3}\sin\theta \tag{13}$$

$$\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\left(\frac{\sigma_0 r_0^3}{3\varepsilon_0 r^2}\cos\theta\right) = 0\,. \tag{14}$$

Note that the result in Eq. (14) is 0 because the function is independent of the azimuthal angle ϕ.

Substituting the results of Eqs. (12), (13), and (14) into the expression of the electric field in Eq. (11) yields:

$$\mathbf{E}(\mathbf{r}) = \frac{2\sigma_0 r_0^3}{3\varepsilon_0 r^3}\cos\theta\,\hat{\mathbf{e}}_r + \frac{\sigma_0 r_0^3}{3\varepsilon_0 r^3}\sin\theta\,\hat{\mathbf{e}}_\theta$$

Taking the common factor, the result is:

$$\mathbf{E}(\mathbf{r}) = \frac{\sigma_0 r_0^3}{3\varepsilon_0 r^3}\left(2\cos\theta\,\hat{\mathbf{e}}_r + \sin\theta\,\hat{\mathbf{e}}_\theta\right) \tag{15}$$

5.3 Electrostatic Field Throughout Space

While the electric potential is continuous across the surface, the electric field exhibits a discontinuity associated with its normal component. We therefore consider only the regions strictly inside and outside the sphere for this vector function. Combining Eqs. (10) and (15), we finally obtain the electrostatic field of the sphere:

$$\mathbf{E}(\mathbf{r}) = \begin{cases} -\dfrac{\sigma_0}{3\varepsilon_0}\,\hat{\mathbf{k}} & \text{if} \quad r < r_0 \\ \dfrac{\sigma_0 r_0^3}{3\varepsilon_0 r^3}\left(2\cos\theta\,\hat{\mathbf{e}}_r + \sin\theta\,\hat{\mathbf{e}}_\theta\right) & \text{if} \quad r > r_0\,. \end{cases}$$

6 Conclusion

Both the electrostatic potential and field throughout space are determined via this analytical approach, which replaces standard integration procedures with a more straightforward formulation. Spherical harmonics are the optimal choice for treating this problem by virtue of the symmetry of the physical system. The derivation reflects the standard method for obtaining the electric field, since calculating a scalar potential is generally much simpler than integrating a vector field. Therefore, it is more efficient to first evaluate the potential and then, determine the electrostatic field through its negative gradient.

Suggested Reading

G. B. Arfken, H. J. Weber, and F. E. Harris. *Mathematical Methods for Physicists*.

J. D. Jackson. *Classical Electrodynamics*.

Version History

v1.4 (June 2026): Typesetting adjustments.

v1.3 (May 2026): Improved gradient operator representation in spherical coordinates for mathematical rigor.

v1.2 (May 2026): Corrected the DOI URL.

v1.1 (May 2026): Minor wording refinements for improved clarity and academic tone. No changes to results or methodology.

v1.0 (April 2026): Original release.